

RUSSIAN MISSION, AK, USA

WMO: 703053

Lat: 61.775N

Lon: 161.319W

Elev: 16

StdP: 101.14

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 26654

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.8	-33.2	-35.8	0.1	-32.9	-33.6	0.2	-30.4	10.3	-8.5	8.7	-10.3	0.8	250	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	23.9	15.6	21.7	14.7	19.7	13.9	16.8	21.8	15.6	20.0	14.7	18.5	1.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	10.5	18.1	13.7	9.8	17.0	12.8	9.2	15.8	46.8	21.8	43.6	19.9	40.9	18.4	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.1	7.2	6.0	-38.8	27.1	4.7	2.3	-42.2	28.8	-44.9	30.1	-47.6	31.4	-51.0	33.1
			-34.8	18.5	1.9	2.0	-36.1	19.9	-37.3	21.0	-38.3	22.1	-39.7	23.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-15.1	-11.3	-10.6	-1.5	6.4	12.3	13.8	12.6	8.1	0.7	-9.3	-13.0
	DBStd	12.31	10.91	8.63	7.26	6.14	4.29	3.07	3.05	2.61	3.34	4.65	7.12	8.57
	HDD10.0	4162	779	598	639	345	126	11	3	7	72	288	580	714
	HDD18.3	6894	1037	831	897	595	370	182	146	179	307	547	830	973
	CDD10.0	317	0	0	0	0	14	81	121	87	15	0	0	0
	CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0	0
	CDH23.3	82	0	0	0	0	2	24	52	3	0	0	0	0
	CDH26.7	10	0	0	0	0	0	3	7	0	0	0	0	0
Wind	WSAvg	2.3	2.6	2.7	2.4	2.3	2.2	2.1	2.2	2.0	2.2	2.5	2.4	2.4
Precipitation	PrecAvg	618	43	38	34	32	32	46	68	80	75	62	60	49
	PrecMax	767	93	112	125	68	73	80	120	142	142	130	138	108
	PrecMin	466	12	7	8	9	10	16	25	28	19	10	3	20
	PrecStd	84	21	25	24	15	15	13	21	24	29	28	33	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	3.9	5.6	14.0	23.1	26.6	27.8	23.5	18.7	11.5	5.3	4.6
		MCWB	2.7	2.1	2.3	6.8	13.2	15.3	18.9	16.1	12.7	8.3	3.9	1.7
	2%	DB	3.5	2.6	3.1	11.2	19.4	23.2	24.6	20.5	15.9	9.8	3.3	3.2
		MCWB	1.8	1.0	0.8	5.6	11.1	14.4	16.9	15.2	11.6	7.4	2.6	1.7
	5%	DB	1.9	1.5	1.5	8.9	16.5	21.0	22.3	18.8	14.1	8.4	1.7	1.2
		MCWB	0.7	0.4	0.0	4.4	9.9	13.5	15.8	14.4	11.0	6.8	1.0	0.5
	10%	DB	-0.3	0.3	0.1	6.9	14.2	19.0	20.0	17.2	12.8	7.2	0.1	-0.5
		MCWB	-1.0	-0.5	-1.1	3.2	8.4	12.7	14.9	13.7	10.4	5.9	-0.5	-1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.4	2.7	7.1	13.8	16.8	20.0	17.3	13.5	9.0	4.1	2.8
		MCDB	4.6	3.6	5.1	13.1	21.3	23.7	26.2	21.4	16.3	10.7	4.6	4.0
	2%	WB	2.0	1.3	1.2	5.7	11.8	15.4	17.6	15.9	12.3	7.9	2.6	1.5
		MCDB	3.5	2.2	2.6	10.7	18.9	21.5	23.2	18.9	14.7	9.2	3.4	2.7
	5%	WB	0.9	0.5	0.3	4.6	10.2	14.2	16.3	15.0	11.5	7.1	1.1	0.5
		MCDB	2.0	1.4	1.5	8.7	15.6	19.5	21.2	17.9	13.7	8.3	1.6	1.3
	10%	WB	-0.8	-0.4	-1.1	3.4	8.9	13.1	15.1	14.2	10.7	5.9	-0.5	-1.0
		MCDB	0.0	0.4	0.1	6.5	13.6	18.2	19.4	16.7	12.6	7.2	0.0	-0.2
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.6	10.6	9.8	10.0	9.3	7.6	6.9	6.6	5.3	6.4	7.2
		MCDBR	6.6	6.7	9.1	12.4	14.5	13.5	13.0	10.0	9.0	5.7	5.1	6.3
		MCWBR	5.7	6.0	7.3	7.6	7.5	6.4	6.4	5.3	5.5	4.2	4.9	5.6
	5% WB	MCDBR	6.3	6.5	8.0	11.9	13.3	11.9	11.6	8.3	7.8	5.2	5.3	6.2
		MCWBR	5.4	6.1	6.6	7.4	6.9	5.7	5.9	4.6	5.1	4.0	5.2	5.7
Clear-Sky Solar Irradiance	taub		0.220	0.272	0.279	0.350	0.369	0.360	0.367	0.379	0.340	0.292	0.230	0.199
	taud		2.042	2.182	2.242	2.107	2.307	2.409	2.428	2.384	2.452	2.416	2.053	1.892
	Ebn at Noon		483	696	841	828	849	866	848	801	771	684	449	290
	Edh at Noon		37	66	89	126	113	104	100	96	75	53	35	24
All-Sky Solar Radiation	RadAvg		0.32	1.01	2.89	4.58	5.25	5.38	4.43	3.46	2.35	1.12	0.38	0.15
	RadStd		0.05	0.13	0.31	0.40	0.49	0.50	0.40	0.38	0.25	0.09	0.06	0.02

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (10 neighbors)	N/A	+1.67	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page